

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |
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| 2(a)(i)   | more, starch / amylose, is produced so iodine (solution), becomes dark(er) / changes from yellow-brown to blue-black ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>1</b> |
| 2(a)(ii)  | <p><i>prediction</i><br/> absorbance remains, constant / reaches plateau / at 1.80 ;<br/> <b>A</b> 1.79–1.81<br/> <b>A</b> increases and flattens</p> <p><i>explanation</i><br/> all, substrate / G-1-P, used up / converted to amylose ;<br/> <b>A</b> colorimeter reached its limit for absorbance (at about 2.00)<br/> <b>A</b> substrate concentration is limiting</p> <p><b>or</b></p> <p><i>prediction</i><br/> absorbance increases less steeply / increase slows down ;</p> <p><i>explanation</i><br/> substrate concentration is (becoming) limiting / AW ;<br/> <b>A</b> substrate concentration is a limiting factor</p> | <b>2</b> |
| 2(a)(iii) | <p>any <b>one</b> from:</p> <p>colorimeter gives, quantitative / numerical, results / readings ;<br/> <i>idea that</i> can use values from the colorimeter to plot graph(s) ;<br/> results are not subjective / no judgements made by eye / AW ;<br/> <i>idea that</i> can use calibration curve to determine concentrations (of starch) ;<br/> can take continuous readings / do not have to take samples ;</p> <p><b>I</b> 'more accurate', 'actual results', 'reliable', 'exact', 'precise', etc.</p>                                                                                                                            | <b>1</b> |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks |
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| 2(b)     | <p>any <b>four</b> from:</p> <ol style="list-style-type: none"> <li>1 substrate / glucose / G 1-P, binds to active site / forms enzyme-substrate complex / forms ESC ;</li> <li>2 <u>end of amylose</u> / glucose (residue) at end of amylose (molecule), binds to active site / forms enzyme-substrate complex / forms ESC ;</li> <li>3 active site changes shape when substrate(s) bind ; <i>ref. to induced fit</i></li> <li>4 enzyme decreases the activation energy ;</li> <li>5 (<math>\alpha</math>-1,4-)<u>glycosidic bond</u> forms ;<br/>A glucosidic</li> <li>6 condensation (reaction) / water formed / AW ;</li> <li>7 AVP ; e.g. any detail of binding to active site – formation of hydrogen bonds<br/>e.g. phosphate ion leaves active site<br/>e.g. substrate molecules, put under strain / reach transition state / AW</li> </ol> <p><i>accept a diagram to show bond formation</i></p> | 4     |

| Question | Answer | Marks |
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